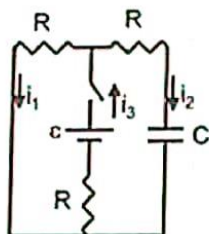


4) A parallel plate capacitor is charged using a battery until the energy stored on the capacitor is 1 J. The battery is then disconnected. If the plates of the capacitor are pulled apart so that the distance between them is doubled, what happens to the energy stored on the capacitor?

- A) The energy stored is now 0.5 J. B) The energy stored is still 1 J.
C) The energy stored is now 2 J. D) The energy stored is now 4 J.

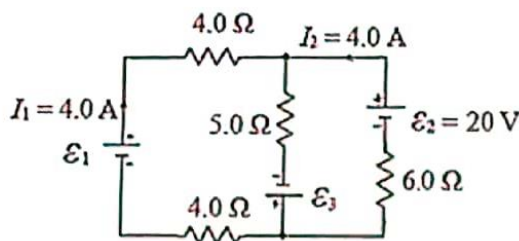
5) Just after the switch in the circuit shown below is closed, the current i_3 (through the battery) is given by

- A) $\varepsilon/3R$
B) $\varepsilon/2R$
C) $3\varepsilon/2R$
D) $2\varepsilon/3R$



6) Consider the circuit shown in the figure. Note that two currents are shown, and that while some of the battery EMFs are given, two (ε_1 and ε_3) are not. What is the EMF ε_1 ?

- A) 12 V
B) 20 V
C) 28 V
D) 44 V



7) A loop of wire carrying a current of 2.0 A is in the shape of a right triangle with two equal sides, each 15 cm long. A 0.7 T uniform magnetic field is in the plane of the triangle and is perpendicular to the hypotenuse (斜边). The resultant magnetic force on the two equal sides has a magnitude of:

- A) 0 N B) 0.21 N C) 0.30 N D) 0.41 N

8) A loop of current-carrying wire has a magnetic dipole moment of $5.0 \times 10^{-4} \text{ A} \cdot \text{m}^2$. If the dipole moment makes an angle of 57° with a magnetic field of 0.35 T, what is its potential energy?

- A) $-9.5 \times 10^{-5} \text{ J}$ B) $-1.5 \times 10^{-4} \text{ J}$ C) $-1.8 \times 10^{-4} \text{ J}$ D) $+1.5 \times 10^{-4} \text{ J}$

9) A solenoid is 3.0 cm long and has a radius of 0.50 cm. It is wrapped with 500 turns of wire carrying a current of 2.0 A. The magnetic field at the center of the solenoid is:

- A) $9.9 \times 10^{-8} \text{ T}$ B) $1.3 \times 10^{-3} \text{ T}$ C) $4.2 \times 10^{-2} \text{ T}$ D) 16 T

10) The magnetic field at any point in the xy plane is given by $\vec{B} = A\vec{r} \times \hat{k}$, where $\vec{r}$ is the position vector of the point, A is a constant, and $\hat{k}$ is a unit vector in the $+z$ direction. The net current through a circle of radius R , in the xy plane and centered at the origin, is given by:

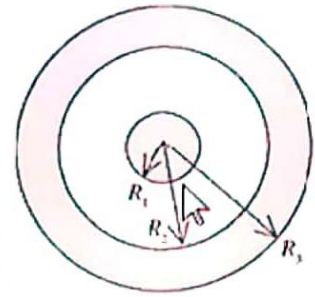
- A) $\pi AR^2/\mu_0$ B) $2\pi AR/\mu_0$ C) $4\pi AR^3/3\mu_0$ D) $2\pi AR^2/\mu_0$

Q2. Blank-filling Questions. (Just write down the result, and the detailed process is not needed.)

1. (3 points) Two concentric spheres are shown in the figure. The inner sphere is a solid nonconductor and has a positive uniform volume charge density $\frac{15}{2\pi} \mu\text{C}/\text{m}^3$. The outer sphere is a conducting shell that carries a net

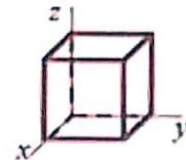
charge of $Q = -15.00 \text{ nC}$. No other charges are present. The radii shown in the figure have the value $R_1 = 10.0 \text{ cm}$, $R_2 = 20.0 \text{ cm}$, and $R_3 = 30.0 \text{ cm}$. Find the potential at $r = 5.0 \text{ cm}$ from the center of inner sphere.

The infinity is defined as potential zero.

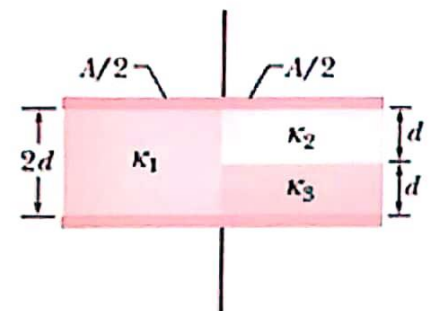


2. (3 points) The figure below shows a cube with sides which are 3.00 m long. One corner of the cube is at the origin, and one side is along each of the Cartesian coordinate axes. Find the net charge enclosed by the cube if there is a non-uniform electric field in this region of space given by

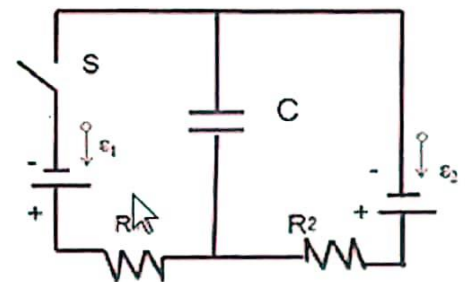
$$\vec{E} = (\sin(30^\circ) \hat{i} + \sin(60^\circ)\cos(30^\circ) \hat{j} + \cos(60^\circ)\cos(30^\circ) \hat{k})(5.00 \text{ x N/Cm})$$

**Text Questions:** (Please write down the detailed process.)

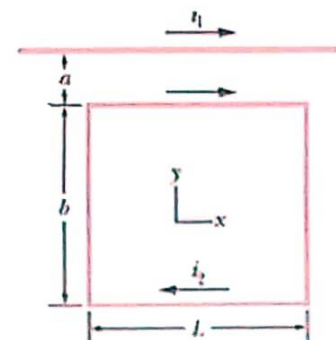
Q3. (10 points) Figure shows a parallel-plate capacitor of plate area $A = 10.5 \text{ cm}^2$ and plate separation $2d = 7.12 \text{ mm}$. The left half of the gap is filled with material of dielectric constant $\kappa_1 = 21.0$; the top of the right half is filled with material of dielectric constant $\kappa_2 = 42.0$; the bottom of the right half is filled with material of dielectric constant $\kappa_3 = 58.0$. What is the capacitance?



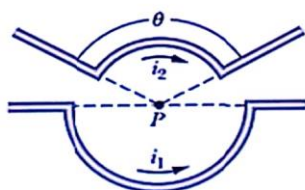
Q4. (10 points) The circuit of figure shows a capacitor, two ideal batteries, two resistors, and a switch S, initially S has been open for a long time. If it is then closed for a long time, what is the change of the potential energy of the capacitor? Assume $C = 15 \mu\text{F}$, $\varepsilon_1 = 1.0 \text{ V}$, $\varepsilon_2 = 3.0 \text{ V}$, $R_1 = 0.30 \Omega$, $R_2 = 0.40 \Omega$



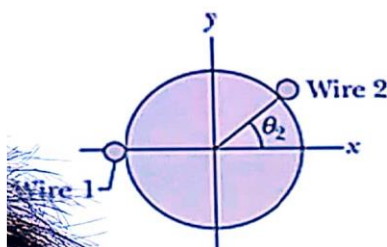
Q5. (10 points) In figure, a long straight wire carries a current $i_1 = 30.0 \text{ A}$ and a rectangular loop carries current $i_2 = 20.0 \text{ A}$. Take $a = 1.00 \text{ cm}$, $b = 8.00 \text{ cm}$, and $L = 30.0 \text{ cm}$. What is the net force on the loop due to i_1 ?



Q6. (10 points) Figure shows two current segments. The lower segment carries a current of $i_1 = 0.40$ A and includes a semicircular arc with radius 5.0 cm, angle 180° , and center point P . The upper segment carries current $i_2 = 2i_1$ and includes a circular arc with radius 4.0 cm, angle 120° , and the same center point P . What is the magnitude of the net magnetic field at P for the indicated current directions?



Q7. (12 points) Figure shows, in cross section, two long straight wires held against a plastic cylinder of radius 20.0 cm. Wire 1 carries current $i_1 = 50.0$ mA out of the page and is fixed in place at the left side of the cylinder. Wire 2 carries current $i_2 = 30.0$ mA out of the page and can be moved around the cylinder. At what (positive) angle θ_2 should wire 2 be positioned such that, at the origin, the net magnetic field due to the two currents has magnitude 43.59 nT?



Q8. (12 points) Your physics assignment is to figure out a way to use electricity to launch a small 6.0 cm long plastic drink stirrer. You decide that you'll charge the little plastic rod by rubbing it with fur, then hold it near an infinite long, charged wire, as shown in figure. When you let go, the electric force of the wire on the plastic rod will shoot it away. Suppose you can uniformly charge the plastic stirrer to 10 nC and that the linear charge density of the long infinite wire is 1.0×10^{-7} C/m. What is the net electric force on the plastic stirrer if the end closest to the wire is 2.0 cm away?

